

Math 1522 Calculus 2 Review: Selected Solutions
Fall 2026

1. Memorization

- (a) Derivative and graph of $\sin^{-1}(x)$.

Solution. $y = \sin^{-1}(x)$ is the inverse of $\sin$ restricted to $[-\frac{\pi}{2}, \frac{\pi}{2}]$, so its domain is $[-1, 1]$ and its range is $[-\frac{\pi}{2}, \frac{\pi}{2}]$. The graph increases from $(-1, -\frac{\pi}{2})$ through $(0, 0)$ to $(1, \frac{\pi}{2})$, with vertical tangents at the ends.

$$\frac{d}{dx} \sin^{-1}(x) = \frac{1}{\sqrt{1-x^2}}, \quad -1 < x < 1.$$

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- (h) State the Inverse Function Theorem.

Solution. Let f be a 1-1 differentiable function with inverse f^{-1} , and let a be in the domain of f^{-1} . If $f'(f^{-1}(a)) \neq 0$, then f^{-1} is differentiable at a and

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}.$$

The tangent line to f at (b, a) , where $b = f^{-1}(a)$, reflected across $y = x$ is the tangent line to f^{-1} at (a, b) , and reflecting swaps rise and run.

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2. Logarithmic Differentiation

- (a) Find y' if $y = x^{\sqrt{x}}$.

Solution. Take $\ln$ of both sides: $\ln y = \sqrt{x} \ln x$. Differentiate implicitly.

$$\frac{y'}{y} = \frac{\ln x}{2\sqrt{x}} + \frac{\sqrt{x}}{x}, \quad y' = x^{\sqrt{x}} \left(\frac{\ln x}{2\sqrt{x}} + \frac{1}{\sqrt{x}} \right) = x^{\sqrt{x}} \frac{\ln x + 2}{2\sqrt{x}}.$$

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- (c) Find y' if $y = (\sin x)^{\sin x}$.

Solution. $\ln y = \sin x \ln(\sin x)$, so

$$\frac{y'}{y} = \cos x \ln(\sin x) + \sin x \cdot \frac{\cos x}{\sin x} = \cos x (\ln(\sin x) + 1),$$

$$y' = (\sin x)^{\sin x} \cos x (\ln(\sin x) + 1).$$

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- (d) Find the slope of the tangent line to $y = x^{\ln x}$ at $x = e$.

Solution. $\ln y = (\ln x)^2$, so $\frac{y'}{y} = \frac{2 \ln x}{x}$ and $y' = x^{\ln x} \cdot \frac{2 \ln x}{x}$. At $x = e$, $y = e^1 = e$, so

$$y'(e) = e \cdot \frac{2}{e} = 2.$$

■

3. Derivatives and Logs

- (c) Find y if $y' = \frac{\sin(x)}{\cos(x)}$.

Solution. Let $u = \cos x$, so $du = -\sin x \, dx$.

$$y = \int \frac{\sin x}{\cos x} \, dx = - \int \frac{du}{u} = -\ln |\cos x| + C = \ln |\sec x| + C.$$

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- (d) Find the domain and range of $f(x) = \ln(\sqrt{9 - e^x})$.

Solution. We need $9 - e^x > 0$, that is $x < \ln 9$. So the domain is $(-\infty, \ln 9)$. On this domain e^x takes every value in $(0, 9)$, so $\sqrt{9 - e^x}$ takes every value in $(0, 3)$, and $\ln$ of that takes every value in $(-\infty, \ln 3)$. The range is $(-\infty, \ln 3)$.

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- (f) Let $f(x) = \sqrt{4 - e^{2x}}$. Find the domain and range of f , find $f^{-1}(x)$, and state the domain and range of f^{-1} . Does the graph of f have a horizontal asymptote?

Solution. Domain: $4 - e^{2x} \geq 0$ means $e^{2x} \leq 4$, that is $x \leq \ln 2$. On $(-\infty, \ln 2]$ the values of e^{2x} fill $(0, 4]$, so $4 - e^{2x}$ fills $[0, 4)$ and the range of f is $[0, 2)$.

Inverse: set $y = \sqrt{4 - e^{2x}}$ and solve for x . Then $y^2 = 4 - e^{2x}$, so $x = \frac{1}{2} \ln(4 - y^2)$. Swapping names,

$$f^{-1}(x) = \frac{1}{2} \ln(4 - x^2),$$

with domain $[0, 2)$ (the range of f) and range $(-\infty, \ln 2]$ (the domain of f).

As $x \rightarrow -\infty$, $e^{2x} \rightarrow 0$ so $f(x) \rightarrow \sqrt{4} = 2$. The line $y = 2$ is a horizontal asymptote.

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- (g) Find $y'(1)$ if $y = 3^x \log_3(x)$.

Solution. Product rule, with $\frac{d}{dx} 3^x = 3^x \ln 3$ and $\frac{d}{dx} \log_3 x = \frac{1}{x \ln 3}$.

$$y' = 3^x \ln 3 \log_3(x) + \frac{3^x}{x \ln 3}, \quad y'(1) = 3 \ln 3 \cdot 0 + \frac{3}{\ln 3} = \frac{3}{\ln 3}.$$

■

4. Derivatives and Inverse trig functions

- (a) Prove the identity $\frac{d}{dx} \sec^{-1}(x) = \frac{1}{x\sqrt{x^2-1}}$ (ignore domain issues).

Solution. Let $y = \sec^{-1}(x)$, so $\sec y = x$. Differentiate implicitly: $\sec y \tan y y' = 1$. From $\tan^2 y = \sec^2 y - 1 = x^2 - 1$ we get $\tan y = \sqrt{x^2 - 1}$ (taking y in $[0, \frac{\pi}{2})$, where $\tan y \geq 0$). So

$$y' = \frac{1}{\sec y \tan y} = \frac{1}{x\sqrt{x^2 - 1}}.$$

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- (d) Find y if $y' = \frac{x^2}{\frac{1}{9} + x^6}$.

Solution. Let $u = x^3$, so $du = 3x^2 dx$.

$$y = \frac{1}{3} \int \frac{du}{\frac{1}{9} + u^2} = \frac{1}{3} \cdot 9 \int \frac{du}{1 + (3u)^2} = \frac{1}{3} \cdot 9 \cdot \frac{1}{3} \tan^{-1}(3u) + C = \tan^{-1}(3x^3) + C.$$

■

- (f) Find the equation of the tangent line to $y = 2 \tan^{-1}(x^2)$ at $x = 1$.

Solution. The point is $y(1) = 2 \tan^{-1}(1) = \frac{\pi}{2}$. The slope is

$$y' = 2 \cdot \frac{2x}{1 + x^4}, \quad y'(1) = 2.$$

So the tangent line is $y = \frac{\pi}{2} + 2(x - 1)$.

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- (g) Write $\cos(\tan^{-1}(x/2))$ as an algebraic function of x .

Solution. Let $\theta = \tan^{-1}(x/2)$, so $\tan \theta = \frac{x}{2}$ with $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. Draw a right triangle with opposite side x and adjacent side 2. The hypotenuse is $\sqrt{4 + x^2}$, and $\cos \theta > 0$ on this range, so

$$\cos(\tan^{-1}(x/2)) = \frac{2}{\sqrt{4 + x^2}}.$$

■

- (h) Find y if $y' = \frac{x}{\sqrt{1-x^4}}$.

Solution. Let $u = x^2$, so $du = 2x dx$.

$$y = \frac{1}{2} \int \frac{du}{\sqrt{1-u^2}} = \frac{1}{2} \sin^{-1}(x^2) + C.$$

■

5. Hyperbolic trig

- (b) Find $\cosh^{-1}(2)$.

Solution. Solve $\cosh y = 2$, that is $\frac{e^y + e^{-y}}{2} = 2$. Multiply by $2e^y$: $e^{2y} - 4e^y + 1 = 0$, a quadratic in e^y . So $e^y = 2 \pm \sqrt{3}$ and $y = \ln(2 \pm \sqrt{3})$. Since $(2 - \sqrt{3})(2 + \sqrt{3}) = 1$, the two solutions are $y = \pm \ln(2 + \sqrt{3})$. The function $\cosh^{-1}$ is the inverse of $\cosh$ restricted to $[0, \infty)$, so

$$\cosh^{-1}(2) = \ln(2 + \sqrt{3}).$$

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(d) Evaluate $\lim_{x \rightarrow \infty} \coth x$.

Solution. Divide the top and bottom by e^x .

$$\lim_{x \rightarrow \infty} \coth x = \lim_{x \rightarrow \infty} \frac{e^x + e^{-x}}{e^x - e^{-x}} = \lim_{x \rightarrow \infty} \frac{1 + e^{-2x}}{1 - e^{-2x}} = \frac{1 + 0}{1 - 0} = 1.$$

■

(f) Prove that the derivative of $\tanh x$ is $\operatorname{sech}^2 x$.

Solution. Write $\tanh x = \frac{\sinh x}{\cosh x}$ and use the quotient rule with $(\sinh x)' = \cosh x$, $(\cosh x)' = \sinh x$.

$$\frac{d}{dx} \tanh x = \frac{\cosh x \cosh x - \sinh x \sinh x}{\cosh^2 x} = \frac{1}{\cosh^2 x} = \operatorname{sech}^2 x,$$

using $\cosh^2 x - \sinh^2 x = 1$.

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(g) Find y' if $y = \tan^{-1}(\sinh x)$, and simplify.

Solution.

$$y' = \frac{\cosh x}{1 + \sinh^2 x} = \frac{\cosh x}{\cosh^2 x} = \frac{1}{\cosh x} = \operatorname{sech} x.$$

■

(h) Evaluate $\int_{-\ln 3}^{\ln 3} \cosh x \, dx$.

Solution.

$$\int_{-\ln 3}^{\ln 3} \cosh x \, dx = \sinh(\ln 3) - \sinh(-\ln 3) = 2 \sinh(\ln 3) = e^{\ln 3} - e^{-\ln 3} = 3 - \frac{1}{3} = \frac{8}{3}.$$

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6. Inverse Function Theorem

(a) Show $f(x) = x + x^3 + 2x^5$ is 1-1, then find the derivative of f^{-1} at $(4, f^{-1}(4))$.

Solution. $f'(x) = 1 + 3x^2 + 10x^4 \geq 1 > 0$, so f is increasing, hence 1-1. Since $f(1) = 1 + 1 + 2 = 4$, we have $f^{-1}(4) = 1$, and

$$(f^{-1})'(4) = \frac{1}{f'(1)} = \frac{1}{14}.$$

■

(b) $f(x) = 5x + 2 \sin x + 2 \cos x$, at $(2, f^{-1}(2))$.

Solution. $f'(x) = 5 + 2 \cos x - 2 \sin x$. Since $|2 \cos x - 2 \sin x| \leq 2\sqrt{2} < 5$, $f'(x) > 0$ and f is 1-1. Since $f(0) = 2$, $f^{-1}(2) = 0$, and

$$(f^{-1})'(2) = \frac{1}{f'(0)} = \frac{1}{5 + 2} = \frac{1}{7}.$$

■

- (i) Let $f(x) = x^3 + x + 1$. Find the equation of the tangent line to the graph of f^{-1} at $x = 3$.

Solution. $f(1) = 3$, so $f^{-1}(3) = 1$ and the point is $(3, 1)$. $f'(x) = 3x^2 + 1$, so $(f^{-1})'(3) = \frac{1}{f'(1)} = \frac{1}{4}$. The tangent line is

$$y = 1 + \frac{1}{4}(x - 3).$$

- (j) Let $f(x) = 2x^3 + \ln x + \sin(\pi x)$ for $x > 0$ (assume f is 1-1). Find the slope of the tangent line to the graph of f^{-1} at $x = 2$.

Solution. $f(1) = 2 + 0 + 0 = 2$, so $f^{-1}(2) = 1$. $f'(x) = 6x^2 + \frac{1}{x} + \pi \cos(\pi x)$, so $f'(1) = 6 + 1 - \pi$. The slope is

$$(f^{-1})'(2) = \frac{1}{f'(1)} = \frac{1}{7 - \pi}.$$

7. L'Hopital's Rule

- (c) $\lim_{x \rightarrow 0^+} x^x$

Solution. This is the form 0^0 . Let $y = x^x$, so $\ln y = x \ln x = \frac{\ln x}{1/x}$, the form $\frac{-\infty}{\infty}$.

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{1/x} \stackrel{H}{=} \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0^+} (-x) = 0.$$

So $\ln y \rightarrow 0$ and $x^x \rightarrow e^0 = 1$.

- (g) $\lim_{x \rightarrow 0^+} (1 + \sin(2x))^{\cot x}$

Solution. The form 1^∞ . Let y be the expression, so

$$\ln y = \cot x \ln(1 + \sin 2x) = \frac{\ln(1 + \sin 2x)}{\tan x}, \text{ the form } \frac{0}{0}.$$

$$\lim_{x \rightarrow 0^+} \frac{\ln(1 + \sin 2x)}{\tan x} \stackrel{H}{=} \lim_{x \rightarrow 0^+} \frac{\frac{2 \cos 2x}{1 + \sin 2x}}{\sec^2 x} = \frac{2}{1} = 2.$$

So the limit is e^2 .

- (i) $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{e^x - 1} \right)$

Solution. The form $\infty - \infty$. Combine into one fraction, then apply L'Hopital twice (each step is $\frac{0}{0}$).

$$\lim_{x \rightarrow 0^+} \frac{e^x - 1 - x}{x(e^x - 1)} \stackrel{H}{=} \lim_{x \rightarrow 0^+} \frac{e^x - 1}{e^x - 1 + xe^x} \stackrel{H}{=} \lim_{x \rightarrow 0^+} \frac{e^x}{2e^x + xe^x} = \frac{1}{2}.$$

(j) $\lim_{x \rightarrow 0} \frac{\tan^{-1}(2x)}{x}$

Solution. The form $\frac{0}{0}$.

$$\lim_{x \rightarrow 0} \frac{\tan^{-1}(2x)}{x} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\frac{2}{1+4x^2}}{1} = 2.$$

(k) $\lim_{x \rightarrow \infty} x \left(\frac{\pi}{2} - \tan^{-1}(2x) \right)$

Solution. The form $\infty \cdot 0$, since $\tan^{-1}(2x) \rightarrow \frac{\pi}{2}$. Rewrite as a quotient of the form $\frac{0}{0}$.

$$\lim_{x \rightarrow \infty} \frac{\frac{\pi}{2} - \tan^{-1}(2x)}{1/x} \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{-\frac{2}{1+4x^2}}{-\frac{1}{x^2}} = \lim_{x \rightarrow \infty} \frac{2x^2}{1+4x^2} = \frac{1}{2}.$$

(l) $\lim_{x \rightarrow \infty} \sin^{-1} \left(\frac{e^x}{2e^x + 1} \right)$

Solution. No L'Hopital needed. Divide by e^x inside: $\frac{e^x}{2e^x + 1} = \frac{1}{2 + e^{-x}} \rightarrow \frac{1}{2}$. Since $\sin^{-1}$ is continuous at $\frac{1}{2}$,

$$\lim_{x \rightarrow \infty} \sin^{-1} \left(\frac{e^x}{2e^x + 1} \right) = \sin^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{6}.$$

(m) $\lim_{x \rightarrow \infty} \frac{\cosh(\ln x)}{x}$

Solution. Use the definition: $\cosh(\ln x) = \frac{e^{\ln x} + e^{-\ln x}}{2} = \frac{x + \frac{1}{x}}{2}$. So

$$\lim_{x \rightarrow \infty} \frac{\cosh(\ln x)}{x} = \lim_{x \rightarrow \infty} \frac{1 + \frac{1}{x^2}}{2} = \frac{1}{2}.$$

8. U-sub

(a) $\int_0^{(\pi/2)^{1/3}} x^2 \sin x^3 dx$

Solution. Let $u = x^3$, $du = 3x^2 dx$. The bounds become $u = 0$ and $u = \frac{\pi}{2}$.

$$\frac{1}{3} \int_0^{\pi/2} \sin u du = \frac{1}{3} \left[-\cos u \right]_0^{\pi/2} = \frac{1}{3} (0 + 1) = \frac{1}{3}.$$

(d) $\int_{\pi/4}^{\pi/2} \cot x dx$

Solution. Let $u = \sin x$, $du = \cos x \, dx$. The bounds become $u = \frac{\sqrt{2}}{2}$ and $u = 1$.

$$\int_{\sqrt{2}/2}^1 \frac{du}{u} = \ln 1 - \ln \frac{\sqrt{2}}{2} = \ln \sqrt{2} = \frac{\ln 2}{2}.$$

(e) $\int_1^e \frac{(\ln x)^2}{x} \, dx$

Solution. Let $u = \ln x$, $du = \frac{dx}{x}$. The bounds become $u = 0$ and $u = 1$.

$$\int_0^1 u^2 \, du = \frac{1}{3}.$$

(f) $\int \frac{\cos x}{4 + \sin^2 x} \, dx$

Solution. Let $u = \sin x$, $du = \cos x \, dx$.

$$\int \frac{du}{4 + u^2} = \frac{1}{2} \tan^{-1} \left(\frac{u}{2} \right) + C = \frac{1}{2} \tan^{-1} \left(\frac{\sin x}{2} \right) + C.$$

(g) $\int \frac{dx}{\sqrt{x}(1+\sqrt{x})}$

Solution. Let $u = 1 + \sqrt{x}$, $du = \frac{dx}{2\sqrt{x}}$.

$$\int \frac{2 \, du}{u} = 2 \ln |u| + C = 2 \ln(1 + \sqrt{x}) + C.$$

9. Integration by Parts

(b) $\int_1^e x^4 \ln(x) \, dx$

Solution. Let $u = \ln x$, $dv = x^4 \, dx$, so $du = \frac{dx}{x}$, $v = \frac{x^5}{5}$.

$$\int x^4 \ln x \, dx = \frac{x^5 \ln x}{5} - \int \frac{x^4}{5} \, dx = \frac{x^5 \ln x}{5} - \frac{x^5}{25} + C.$$

$$\int_1^e x^4 \ln x \, dx = \left(\frac{e^5}{5} - \frac{e^5}{25} \right) - \left(0 - \frac{1}{25} \right) = \frac{4e^5 + 1}{25}.$$

(e) $\int_0^\pi e^x \sin(x) \, dx$

Solution. Call the antiderivative $I = \int e^x \sin x \, dx$. Parts with $u = \sin x$, $dv = e^x \, dx$, then again with $u = \cos x$, $dv = e^x \, dx$:

$$I = e^x \sin x - \int e^x \cos x \, dx = e^x \sin x - e^x \cos x - I.$$

So $I = \frac{e^x(\sin x - \cos x)}{2} + C$, and

$$\int_0^\pi e^x \sin x \, dx = \frac{e^\pi(0+1)}{2} - \frac{1 \cdot (0-1)}{2} = \frac{e^\pi + 1}{2}.$$

■

(g) $\int x \sec^2(x) \, dx$

Solution. Let $u = x$, $dv = \sec^2 x \, dx$, so $du = dx$, $v = \tan x$.

$$\int x \sec^2 x \, dx = x \tan x - \int \tan x \, dx = x \tan x + \ln |\cos x| + C.$$

■

(h) $\int e^x \cos(2x) \, dx$

Solution. Call it I . Parts with $u = \cos 2x$, $dv = e^x \, dx$, then with $u = \sin 2x$, $dv = e^x \, dx$:

$$I = e^x \cos 2x + 2 \int e^x \sin 2x \, dx = e^x \cos 2x + 2e^x \sin 2x - 4I.$$

So $5I = e^x(\cos 2x + 2 \sin 2x)$ and $I = \frac{e^x(\cos 2x + 2 \sin 2x)}{5} + C$.

■

(i) $\int_0^1 x \tan^{-1}(x) \, dx$

Solution. Let $u = \tan^{-1} x$, $dv = x \, dx$, so $du = \frac{dx}{1+x^2}$, $v = \frac{x^2}{2}$.

$$\int x \tan^{-1} x \, dx = \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{x^2}{1+x^2} \, dx = \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} (x - \tan^{-1} x) + C,$$

using $\frac{x^2}{1+x^2} = 1 - \frac{1}{1+x^2}$. At the bounds,

$$\int_0^1 x \tan^{-1} x \, dx = \frac{\pi}{8} - \frac{1}{2} \left(1 - \frac{\pi}{4}\right) = \frac{\pi}{4} - \frac{1}{2}.$$

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10. Trig Integrals

(a) $\int \sin^3(x) \cos^2(x) \, dx$

Solution. The power of sin is odd, so save one $\sin x$ and convert the rest with $\sin^2 x = 1 - \cos^2 x$. Let $u = \cos x$, $du = -\sin x \, dx$.

$$\int (1 - \cos^2 x) \cos^2 x \sin x \, dx = - \int (u^2 - u^4) \, du = -\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + C.$$

■

(f) $\int \tan^2(x) dx$

Solution. Use $\tan^2 x = \sec^2 x - 1$.

$$\int (\sec^2 x - 1) dx = \tan x - x + C.$$

(h) $\int \tan^2(x) \sec^4(x) dx$

Solution. The power of sec is even, so save $\sec^2 x$ and convert the other $\sec^2 x = 1 + \tan^2 x$. Let $u = \tan x$, $du = \sec^2 x dx$.

$$\int \tan^2 x (1 + \tan^2 x) \sec^2 x dx = \int (u^2 + u^4) du = \frac{\tan^3 x}{3} + \frac{\tan^5 x}{5} + C.$$

(i) $\int \sin(3x) \cos(2x) dx$

Solution. Use the product formula $\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$ with $A = 3x$, $B = 2x$.

$$\frac{1}{2} \int (\sin 5x + \sin x) dx = -\frac{\cos 5x}{10} - \frac{\cos x}{2} + C.$$

(j) $\int_0^{\pi/2} \cos^2(3x) dx$

Solution. Use $\cos^2 \theta = \frac{1+\cos 2\theta}{2}$ with $\theta = 3x$.

$$\int_0^{\pi/2} \frac{1 + \cos 6x}{2} dx = \left[\frac{x}{2} + \frac{\sin 6x}{12} \right]_0^{\pi/2} = \frac{\pi}{4} + \frac{\sin 3\pi}{12} = \frac{\pi}{4}.$$

11. Trig sub

(a) $\int_2^3 \frac{dx}{(x^2-1)^{3/2}}$

Solution. Let $x = \sec \theta$, $dx = \sec \theta \tan \theta d\theta$, so $(x^2 - 1)^{3/2} = \tan^3 \theta$ (for $0 \leq \theta < \frac{\pi}{2}$).

$$\int \frac{\sec \theta \tan \theta}{\tan^3 \theta} d\theta = \int \frac{\cos \theta}{\sin^2 \theta} d\theta = -\frac{1}{\sin \theta} + C = -\frac{x}{\sqrt{x^2-1}} + C,$$

since $\sin \theta = \frac{\sqrt{x^2-1}}{x}$ from the triangle with hypotenuse x and adjacent side 1.

$$\int_2^3 \frac{dx}{(x^2-1)^{3/2}} = -\frac{3}{\sqrt{8}} + \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3} - \frac{3\sqrt{2}}{4}.$$

(g) $\int \frac{dx}{(6-8x-x^2)^{3/2}}$

Solution. Complete the square: $6 - 8x - x^2 = 22 - (x + 4)^2$. Let $x + 4 = \sqrt{22} \sin \theta$, $dx = \sqrt{22} \cos \theta d\theta$, so $22 - (x + 4)^2 = 22 \cos^2 \theta$.

$$\int \frac{\sqrt{22} \cos \theta}{22^{3/2} \cos^3 \theta} d\theta = \frac{1}{22} \int \sec^2 \theta d\theta = \frac{\tan \theta}{22} + C = \frac{x + 4}{22\sqrt{6 - 8x - x^2}} + C.$$

■

(h) $\int_0^{\sqrt{3}} \frac{x^3}{\sqrt{x^2+1}} dx$

Solution. Let $x = \tan \theta$, $dx = \sec^2 \theta d\theta$, so $\sqrt{x^2 + 1} = \sec \theta$. The bounds become $\theta = 0$ and $\theta = \frac{\pi}{3}$.

$$\int_0^{\pi/3} \tan^3 \theta \sec \theta d\theta = \int_0^{\pi/3} (\sec^2 \theta - 1) \sec \theta \tan \theta d\theta.$$

Let $w = \sec \theta$, $dw = \sec \theta \tan \theta d\theta$, with bounds $w = 1$ and $w = 2$.

$$\int_1^2 (w^2 - 1) dw = \left[\frac{w^3}{3} - w \right]_1^2 = \frac{2}{3} + \frac{2}{3} = \frac{4}{3}.$$

■

(i) $\int_{\sqrt{2}}^2 \frac{dx}{x^2 \sqrt{x^2-1}}$

Solution. Let $x = \sec \theta$, $dx = \sec \theta \tan \theta d\theta$, so $\sqrt{x^2 - 1} = \tan \theta$. The bounds become $\theta = \frac{\pi}{4}$ and $\theta = \frac{\pi}{3}$.

$$\int_{\pi/4}^{\pi/3} \frac{\sec \theta \tan \theta}{\sec^2 \theta \tan \theta} d\theta = \int_{\pi/4}^{\pi/3} \cos \theta d\theta = \frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2}.$$

■

(j) $\int_0^1 \frac{dx}{(1+x^2)^2}$

Solution. Let $x = \tan \theta$, $dx = \sec^2 \theta d\theta$, so $(1 + x^2)^2 = \sec^4 \theta$. The bounds become $\theta = 0$ and $\theta = \frac{\pi}{4}$.

$$\int_0^{\pi/4} \cos^2 \theta d\theta = \left[\frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_0^{\pi/4} = \frac{\pi}{8} + \frac{1}{4}.$$

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12. Partial Fraction Decomposition

(a)(ii) $\int \frac{3x^2+6x+2}{x^2+3x+2} dx$

Solution. The degrees are equal, so divide first: $\frac{3x^2+6x+2}{x^2+3x+2} = 3 - \frac{3x+4}{(x+1)(x+2)}$. Write $\frac{3x+4}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$, so $3x + 4 = A(x + 2) + B(x + 1)$. At $x = -1$, $A = 1$. At $x = -2$, $B = 2$.

$$\int \left(3 - \frac{1}{x+1} - \frac{2}{x+2} \right) dx = 3x - \ln |x+1| - 2 \ln |x+2| + C.$$

■

(b)(iv) $\int \frac{x+4}{x^3-4x} dx$

Solution. Factor $x^3 - 4x = x(x-2)(x+2)$ and write $\frac{x+4}{x(x-2)(x+2)} = \frac{A}{x} + \frac{B}{x-2} + \frac{D}{x+2}$ (we use D to keep C for the constant). Then
 $x+4 = A(x-2)(x+2) + Bx(x+2) + Dx(x-2)$. At $x=0$, $4 = -4A$, so $A = -1$. At $x=2$, $6 = 8B$, so $B = \frac{3}{4}$. At $x=-2$, $2 = 8D$, so $D = \frac{1}{4}$.

$$\int \frac{x+4}{x^3-4x} dx = -\ln|x| + \frac{3}{4} \ln|x-2| + \frac{1}{4} \ln|x+2| + C.$$

■

(c)(ii) $\int \frac{3x^2-x+8}{x^3+4x} dx$

Solution. $x^3 + 4x = x(x^2 + 4)$ with $x^2 + 4$ irreducible. Write $\frac{3x^2-x+8}{x(x^2+4)} = \frac{A}{x} + \frac{Bx+D}{x^2+4}$, so
 $3x^2 - x + 8 = A(x^2 + 4) + (Bx + D)x$. At $x=0$, $A=2$. Matching x^2 : $3 = A + B$, so $B=1$. Matching x : $D=-1$.

$$\int \left(\frac{2}{x} + \frac{x}{x^2+4} - \frac{1}{x^2+4} \right) dx = 2 \ln|x| + \frac{1}{2} \ln(x^2+4) - \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + C.$$

■

(c)(iii) Set up, but do not solve, the partial fraction decomposition of $\frac{x+1}{x^2(x-3)(x^2+1)^2}$.

Solution. Each power of a linear factor gets a constant on top, and each power of an irreducible quadratic gets a linear term on top.

$$\frac{x+1}{x^2(x-3)(x^2+1)^2} = \frac{A}{x} + \frac{B}{x^2} + \frac{D}{x-3} + \frac{Ex+F}{x^2+1} + \frac{Gx+H}{(x^2+1)^2}.$$

■

13. Improper Integrals

(b) $\int_0^1 \frac{\ln x}{x} dx$

Solution. The integrand is undefined at 0. With $u = \ln x$, an antiderivative is $\frac{(\ln x)^2}{2}$.

$$\lim_{t \rightarrow 0^+} \int_t^1 \frac{\ln x}{x} dx = \lim_{t \rightarrow 0^+} \left(0 - \frac{(\ln t)^2}{2} \right) = -\infty.$$

It diverges. The region between the graph and the x -axis over $(0, 1]$, below the axis and running down along the y -axis, has infinite area. ■

(f) $\int_1^\infty \frac{dx}{x^p}$ (consider $p \leq 1$ and $p > 1$)

Solution. If $p=1$: $\lim_{t \rightarrow \infty} \ln t = \infty$, divergent. If $p \neq 1$:

$$\lim_{t \rightarrow \infty} \int_1^t x^{-p} dx = \lim_{t \rightarrow \infty} \frac{t^{1-p} - 1}{1-p}.$$

For $p > 1$ the exponent $1-p$ is negative, so $t^{1-p} \rightarrow 0$ and the integral is $\frac{1}{p-1}$. For $p < 1$, $t^{1-p} \rightarrow \infty$ and it diverges. So it converges exactly when $p > 1$. ■

- (l) $\int_0^\infty e^{-x^2} dx$ (do not determine the value)

Solution. Split at 1. The piece $\int_0^1 e^{-x^2} dx$ is an ordinary integral of a continuous function. For $x \geq 1$, $x^2 \geq x$, so $0 < e^{-x^2} \leq e^{-x}$, and

$$\int_1^\infty e^{-x} dx = \lim_{t \rightarrow \infty} (e^{-1} - e^{-t}) = \frac{1}{e}.$$

By comparison $\int_1^\infty e^{-x^2} dx$ converges, so $\int_0^\infty e^{-x^2} dx$ converges. ■

- (n) $\int_0^\infty \frac{e^x}{(1+e^x)^3} dx$

Solution. Let $u = 1 + e^x$, $du = e^x dx$; $x = 0$ gives $u = 2$ and $x \rightarrow \infty$ gives $u \rightarrow \infty$.

$$\lim_{t \rightarrow \infty} \int_2^t u^{-3} du = \lim_{t \rightarrow \infty} \left(\frac{1}{2 \cdot 4} - \frac{1}{2t^2} \right) = \frac{1}{8}.$$

- (o) $\int_1^\infty \frac{\ln x}{x^3} dx$

Solution. Parts with $u = \ln x$, $dv = x^{-3} dx$, so $du = \frac{dx}{x}$, $v = -\frac{1}{2x^2}$.

$$\int \frac{\ln x}{x^3} dx = -\frac{\ln x}{2x^2} + \int \frac{dx}{2x^3} = -\frac{\ln x}{2x^2} - \frac{1}{4x^2} + C.$$

$$\lim_{t \rightarrow \infty} \left(-\frac{\ln t}{2t^2} - \frac{1}{4t^2} + \frac{1}{4} \right) = \frac{1}{4},$$

since $\frac{\ln t}{t^2} \rightarrow 0$ by L'Hopital. ■

- (p) $\int_0^2 \frac{dx}{(x-1)^{2/3}}$

Solution. The integrand blows up at $x = 1$, inside the interval, so split there. An antiderivative is $3(x-1)^{1/3}$.

$$\lim_{t \rightarrow 1^-} \left[3(x-1)^{1/3} \right]_0^t + \lim_{s \rightarrow 1^+} \left[3(x-1)^{1/3} \right]_s^2 = (0+3) + (3-0) = 6.$$

Both pieces converge, so the integral converges to 6. ■

- (q) $\int_2^\infty \frac{dx}{4+x^2}$

Solution.

$$\lim_{t \rightarrow \infty} \left[\frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) \right]_2^t = \frac{1}{2} \cdot \frac{\pi}{2} - \frac{1}{2} \cdot \frac{\pi}{4} = \frac{\pi}{8}.$$

14. Numerical integration

- (c) $\int_0^1 (x^3 - 3x^2 + 2x + 1) dx$: over or under estimates, then M_4 and T_4 .

Solution. Let $f(x) = x^3 - 3x^2 + 2x + 1$. Then $f''(x) = 6x - 6 < 0$ on $[0, 1]$, so f is concave down. A concave down graph lies below its chords and above its tangent lines, so the trapezoid rule underestimates and the midpoint rule overestimates.

$f'(x) = 3x^2 - 6x + 2$ changes sign at $x = 1 - \frac{1}{\sqrt{3}}$, so f is not monotonic on $[0, 1]$, and the left and right endpoint rules are not determined by this reasoning.

With $\Delta x = \frac{1}{4}$, $M_4 = \frac{1}{4}(f(\frac{1}{8}) + f(\frac{3}{8}) + f(\frac{5}{8}) + f(\frac{7}{8})) = \frac{161}{128} \approx 1.258$ and $T_4 = \frac{1}{8}(f(0) + 2f(\frac{1}{4}) + 2f(\frac{1}{2}) + 2f(\frac{3}{4}) + f(1)) = \frac{79}{64} \approx 1.234$. The exact value is $\frac{1}{4} - 1 + 1 + 1 = \frac{5}{4}$, between them as predicted. ■

- (d) Use the trapezoid rule with $n = 3$ to approximate $\int_0^3 2^x dx$ in exact form.

Solution. $\Delta x = 1$ and the nodes are 0, 1, 2, 3.

$$T_3 = \frac{1}{2}(2^0 + 2 \cdot 2^1 + 2 \cdot 2^2 + 2^3) = \frac{1}{2}(1 + 4 + 8 + 8) = \frac{21}{2}.$$

(The exact value is $\frac{7}{\ln 2} \approx 10.10$. 2^x is concave up, so T_3 overestimates.) ■

- (e) How large must n be so that the trapezoid rule approximates $\int_1^2 \frac{dx}{x}$ with error less than 10^{-4} ?

Solution. $f(x) = \frac{1}{x}$ has $f''(x) = \frac{2}{x^3}$, which on $[1, 2]$ is at most 2. So take $K = 2$, $b - a = 1$.

$$|E_T| \leq \frac{2}{12n^2} = \frac{1}{6n^2} < 10^{-4} \iff n^2 > \frac{10^4}{6} \approx 1666.7.$$

So $n \geq 41$ is enough. ■

15. Sequences

- (e) If an alternating sequence converges, what must it converge to?

Solution. It must converge to 0. Suppose $a_n \rightarrow L$ and the terms alternate in sign. The positive terms are a subsequence converging to L , so $L \geq 0$. The negative terms also converge to L , so $L \leq 0$. Hence $L = 0$. ■

- (h) Write $\{-\frac{1}{8}, \frac{1}{27}, -\frac{1}{64}, \frac{1}{125}, \dots\}$ as a function of n .

Solution. The denominators are $2^3, 3^3, 4^3, 5^3$ and the signs alternate starting negative.

So $a_n = \frac{(-1)^n}{(n+1)^3}$ for $n \geq 1$. ■

- (j) Monotonic? Bounded? $a_n = 1 + \frac{(-1)^n}{n}$ and $b_n = \frac{n}{n+1}$ ($n \geq 1$).

Solution. $a_1 = 0$, $a_2 = \frac{3}{2}$, $a_3 = \frac{2}{3}$, $a_4 = \frac{5}{4}$: it goes up, down, up, so a_n is not monotonic.

Since $\left| \frac{(-1)^n}{n} \right| \leq 1$, $0 \leq a_n \leq 2$, so it is bounded.

$b_n = 1 - \frac{1}{n+1}$, and $\frac{1}{n+1}$ decreases, so b_n is increasing. It is bounded: $\frac{1}{2} \leq b_n < 1$. ■

- (k) Show that $\lim_{n \rightarrow \infty} \frac{3^n}{n!} = 0$.

Solution. For $n \geq 3$, write the quotient as a product of n factors and bound the middle ones by 1:

$$0 < \frac{3^n}{n!} = \frac{3}{1} \cdot \frac{3}{2} \cdot \frac{3}{3} \cdot \frac{3}{4} \cdots \frac{3}{n-1} \cdot \frac{3}{n} \leq \frac{3}{1} \cdot \frac{3}{2} \cdot \frac{3}{n} = \frac{27}{2n}.$$

Since $\frac{27}{2n} \rightarrow 0$, the Squeeze Theorem gives $\frac{3^n}{n!} \rightarrow 0$. ■

16. Test for Divergence (TFD)

(a) $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n+1}$

Solution. $\left| \frac{(-1)^n n}{n+1} \right| = \frac{n}{n+1} \rightarrow 1$, so the terms do not go to 0 (they approach 1 and -1 alternately). By the TFD the series diverges. ■

(c) $\sum_{n=0}^{\infty} \frac{n^2+1}{6n^2+6n+1}$

Solution. $\lim_{n \rightarrow \infty} \frac{n^2+1}{6n^2+6n+1} = \frac{1}{6} \neq 0$, so the series diverges by the TFD. ■

(e) Can you use the TFD on $\sum_{n=2}^{\infty} \frac{1}{n^2}$?

Solution. No. The TFD only concludes divergence when $a_n \not\rightarrow 0$. Here $\frac{1}{n^2} \rightarrow 0$, so the TFD says nothing. (The series converges, as a p -series with $p = 2$.) ■

(f) $\sum_{n=1}^{\infty} 2^{1/n}$

Solution. $2^{1/n} \rightarrow 2^0 = 1 \neq 0$, so the series diverges by the TFD. ■

(g) Suppose $b_n > 0$ and $\sum b_n$ converges. What does the TFD say about $\sum \frac{1}{b_n}$?

Solution. Since $\sum b_n$ converges, $b_n \rightarrow 0$. As $b_n > 0$, this means $\frac{1}{b_n} \rightarrow \infty$, so the terms of $\sum \frac{1}{b_n}$ do not go to 0. By the TFD, $\sum \frac{1}{b_n}$ diverges. ■

17. Telescoping Series

(a) $\sum_{n=2}^{\infty} \frac{1}{n^2-1}$

Solution. Partial fractions: $\frac{1}{n^2-1} = \frac{1}{2} \left(\frac{1}{n-1} - \frac{1}{n+1} \right)$. In the partial sum $s_N = \sum_{n=2}^N$, everything cancels except the first two positive terms and the last two negative ones.

$$s_N = \frac{1}{2} \left(1 + \frac{1}{2} - \frac{1}{N} - \frac{1}{N+1} \right) \rightarrow \frac{1}{2} \cdot \frac{3}{2} = \frac{3}{4}.$$

■

(c) $\sum_{n=1}^{\infty} [\tan^{-1}(n+2) - \tan^{-1}(n)]$

Solution. Each term cancels against the term two places later.

$$s_N = \tan^{-1}(N+1) + \tan^{-1}(N+2) - \tan^{-1}(1) - \tan^{-1}(2).$$

As $N \rightarrow \infty$, $s_N \rightarrow \frac{\pi}{2} + \frac{\pi}{2} - \frac{\pi}{4} - \tan^{-1}(2) = \frac{3\pi}{4} - \tan^{-1}(2)$. ■

(d) $\sum_{n=1}^{\infty} [f(n) - f(n+1)]$, where $\lim_{n \rightarrow \infty} f(n) = L$

Solution.

$s_N = (f(1) - f(2)) + (f(2) - f(3)) + \cdots + (f(N) - f(N+1)) = f(1) - f(N+1)$. So the sum is $f(1) - L$. ■

(e) $\sum_{n=1}^{\infty} [f(n) - f(n+k)]$, k a positive integer, $\lim f(n) = L$

Solution. For $N \geq k$, every $f(m)$ with $k < m \leq N$ appears once with each sign, so

$$s_N = (f(1) + \cdots + f(k)) - (f(N+1) + \cdots + f(N+k)) \rightarrow f(1) + \cdots + f(k) - kL.$$

■

(f) $\sum_{k=3}^{\infty} \frac{4}{(k-1)(k+1)}$

Solution. $\frac{4}{(k-1)(k+1)} = 2 \left(\frac{1}{k-1} - \frac{1}{k+1} \right)$. In $s_N = \sum_{k=3}^N$ the surviving terms are

$$s_N = 2 \left(\frac{1}{2} + \frac{1}{3} - \frac{1}{N} - \frac{1}{N+1} \right) \rightarrow 2 \cdot \frac{5}{6} = \frac{5}{3}.$$

■

18. Geometric Series

(b) $\sum_{n=0}^{\infty} \frac{4^{2n-1}}{17^n}$

Solution. $\frac{4^{2n-1}}{17^n} = \frac{1}{4} \left(\frac{16}{17} \right)^n$, geometric with first term $\frac{1}{4}$ and ratio $r = \frac{16}{17}$, $|r| < 1$.

$$\sum_{n=0}^{\infty} \frac{1}{4} \left(\frac{16}{17} \right)^n = \frac{1/4}{1 - 16/17} = \frac{17}{4}.$$

■

(c) $\sum_{n=-4}^{\infty} \left(\frac{1}{2} \right)^n$

Solution. The first term is $\left(\frac{1}{2} \right)^{-4} = 16$ and $r = \frac{1}{2}$. So the sum is $\frac{16}{1-1/2} = 32$. ■

(d) $\sum_{n=1}^{\infty} \frac{3^{n+1}}{4^{n-1}}$

Solution. The first term ($n = 1$) is $\frac{9}{1} = 9$ and the ratio is $r = \frac{3}{4}$.

$$\sum_{n=1}^{\infty} \frac{3^{n+1}}{4^{n-1}} = \frac{9}{1 - 3/4} = 36.$$

■

(e) $2 - 1 + \frac{1}{2} - \frac{1}{4} + \frac{1}{8} - \cdots$

Solution. Geometric with first term 2 and $r = -\frac{1}{2}$. The sum is $\frac{2}{1+1/2} = \frac{4}{3}$. ■

- (f) For which a does $\sum_{k=1}^{\infty} \left(\frac{2a}{1+a^2}\right)^k$ converge? Find its value.

Solution. Let $r = \frac{2a}{1+a^2}$. The series converges exactly when $|r| < 1$. Since $(|a| - 1)^2 \geq 0$, we get $2|a| \leq 1 + a^2$, so $|r| \leq 1$, with equality exactly when $|a| = 1$. So the series converges for all $a \neq \pm 1$. Its first term is r , so

$$\sum_{k=1}^{\infty} r^k = \frac{r}{1-r} = \frac{2a}{1+a^2-2a} = \frac{2a}{(a-1)^2}.$$

■

19. Integral Test

- (a) $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$

Solution. $f(x) = \frac{1}{x(\ln x)^2}$ is positive, continuous, and decreasing on $[2, \infty)$. With $u = \ln x$,

$$\int_2^{\infty} \frac{dx}{x(\ln x)^2} = \lim_{t \rightarrow \infty} \left(\frac{1}{\ln 2} - \frac{1}{\ln t} \right) = \frac{1}{\ln 2}.$$

The integral converges, so the series converges.

■

- (c) Show $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if $p > 1$ and diverges if $0 < p \leq 1$.

Solution. For $p > 0$, $f(x) = x^{-p}$ is positive, continuous, and decreasing on $[1, \infty)$. By problem 13(f), $\int_1^{\infty} x^{-p} dx$ converges exactly when $p > 1$. By the Integral Test the series converges for $p > 1$ and diverges for $0 < p \leq 1$, including $p = 1$, where $\int_1^{\infty} \frac{dx}{x} = \lim_{t \rightarrow \infty} \ln t = \infty$.

■

- (d) $\sum_{n=1}^{\infty} n e^{-n^2}$

Solution. $f(x) = x e^{-x^2}$ is positive and continuous, and $f'(x) = e^{-x^2}(1 - 2x^2) < 0$ for $x \geq 1$, so it is decreasing on $[1, \infty)$. With $u = x^2$,

$$\int_1^{\infty} x e^{-x^2} dx = \lim_{t \rightarrow \infty} \frac{e^{-1} - e^{-t^2}}{2} = \frac{1}{2e}.$$

The integral converges, so the series converges.

■

- (e) Can you use the integral test on $\sum_{n=1}^{\infty} \frac{2+\sin n}{n^2}$?

Solution. No. The Integral Test needs $f(x) = \frac{2+\sin x}{x^2}$ to be decreasing, and it is not: the numerator oscillates, so f increases on stretches where $\sin x$ rises quickly. Use the DCT instead: $0 < \frac{2+\sin n}{n^2} \leq \frac{3}{n^2}$, and $\sum \frac{3}{n^2}$ converges ($p = 2$). So the series converges.

■

20. p-series

- (a) $\sum_{n=5}^{\infty} \frac{1}{n^e}$

Solution. A p -series with $p = e > 1$ (starting at $n = 5$ does not change convergence). It converges.

■

(c) $\sum_{n=1}^{\infty} \frac{1}{n^{0.000001}}$

Solution. A p -series with $p = 10^{-6} \leq 1$. It diverges, no matter how slowly. ■

(d) $1 + \frac{1}{2\sqrt{2}} + \frac{1}{3\sqrt{3}} + \frac{1}{4\sqrt{4}} + \cdots$

Solution. The n th term is $\frac{1}{n\sqrt{n}} = \frac{1}{n^{3/2}}$. A p -series with $p = \frac{3}{2} > 1$, so it converges. ■

(e) For which p does $\sum_{n=1}^{\infty} \frac{1}{n^{2p-1}}$ converge?

Solution. It is a p -series with exponent $2p - 1$, so it converges exactly when $2p - 1 > 1$, that is $p > 1$. ■

21. Direct Comparison Test (DCT) and Limit Comparison Test (LCT)

(b) $\sum_{n=1}^{\infty} \sin(1/n)$

Solution. The terms are positive. Compare with $b_n = \frac{1}{n}$, using $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$:

$$\lim_{n \rightarrow \infty} \frac{\sin(1/n)}{1/n} = 1.$$

The limit is positive and finite and $\sum \frac{1}{n}$ diverges, so by the LCT the series diverges. ■

(c) $\sum_{n=0}^{\infty} \frac{\tan^{-1}(n)}{n^2+1}$

Solution. $0 \leq \tan^{-1}(n) < \frac{\pi}{2}$, so for $n \geq 1$, $0 \leq \frac{\tan^{-1} n}{n^2+1} \leq \frac{\pi/2}{n^2}$. Since $\sum \frac{\pi/2}{n^2}$ converges ($p = 2$), the DCT shows the series converges. (The $n = 0$ term is 0 and does not matter.) ■

(f) $\sum_{n=1}^{\infty} \frac{\cos^2 n}{n^2+n}$

Solution. $0 \leq \frac{\cos^2 n}{n^2+n} \leq \frac{1}{n^2}$, and $\sum \frac{1}{n^2}$ converges. By the DCT the series converges. ■

(g) $\sum_{n=1}^{\infty} \frac{n^2-2n}{n^{7/2}+1}$

Solution. The terms are positive for $n \geq 3$, and the first two terms do not affect convergence. For large n the term behaves like $\frac{n^2}{n^{7/2}} = \frac{1}{n^{3/2}}$, so compare with $b_n = \frac{1}{n^{3/2}}$.

$$\lim_{n \rightarrow \infty} \frac{(n^2 - 2n)n^{3/2}}{n^{7/2} + 1} = \lim_{n \rightarrow \infty} \frac{1 - \frac{2}{n}}{1 + n^{-7/2}} = 1.$$

Since $\sum \frac{1}{n^{3/2}}$ converges, the LCT shows the series converges. ■

(h) $\sum_{n=1}^{\infty} \frac{1}{n^2 + n \cos n}$

Solution. $n^2 + n \cos n \geq n^2 - n \geq 0$, with $1 + \cos 1 > 0$ at $n = 1$, so the terms are positive. Compare with $b_n = \frac{1}{n^2}$:

$$\lim_{n \rightarrow \infty} \frac{n^2}{n^2 + n \cos n} = \lim_{n \rightarrow \infty} \frac{1}{1 + \frac{\cos n}{n}} = 1,$$

since $\left| \frac{\cos n}{n} \right| \leq \frac{1}{n} \rightarrow 0$. By the LCT the series converges. ■

22. Alternating Series Test (AST)

(b) $\sum_{i=1}^{\infty} \frac{\cos(\pi i)}{i^{0.1}}$

Solution. $\cos(\pi i) = (-1)^i$, so this is $\sum (-1)^i b_i$ with $b_i = \frac{1}{i^{0.1}}$. Then $b_i > 0$, b_i is decreasing, and $b_i \rightarrow 0$. By the AST it converges. ■

(c) $\sum_{k=2}^{\infty} \frac{(-1)^{k+15}}{\ln k}$

Solution. $(-1)^{k+15} = -(-1)^k$, so this is $-\sum (-1)^k b_k$ with $b_k = \frac{1}{\ln k}$. Since $\ln k$ increases to ∞ , b_k is positive, decreasing, and $b_k \rightarrow 0$. By the AST it converges. ■

(d) $\sum_{n=1}^{\infty} \frac{(-1)^n \ln n}{\sqrt{n}}$

Solution. Let $b_n = \frac{\ln n}{\sqrt{n}}$ and $f(x) = \frac{\ln x}{\sqrt{x}}$. Then $f'(x) = \frac{2-\ln x}{2x^{3/2}} < 0$ for $x > e^2 \approx 7.4$, so b_n is decreasing for $n \geq 8$. By L'Hopital, $\lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt{x}} = \lim_{x \rightarrow \infty} \frac{1/x}{1/(2\sqrt{x})} = \lim_{x \rightarrow \infty} \frac{2}{\sqrt{x}} = 0$. The AST applies from $n = 8$ on, and the first seven terms do not affect convergence. It converges. ■

(e) $\sum_{n=1}^{\infty} \frac{(-1)^n n^2}{n^2+1}$. Can you use the AST?

Solution. No. Here $b_n = \frac{n^2}{n^2+1} \rightarrow 1$, so the hypothesis $b_n \rightarrow 0$ fails. In fact the terms $(-1)^n b_n$ do not go to 0, so the series diverges by the TFD. ■

(f) $a_n = \frac{1}{n}$ for n odd, $a_n = \frac{1}{n^2}$ for n even. Can you use the AST on $\sum (-1)^n a_n$?

Solution. No. The AST needs a_n decreasing, but $a_2 = \frac{1}{4} < a_3 = \frac{1}{3}$, and this happens at every even n . (The series actually diverges. Its even terms form the convergent series $\sum \frac{1}{(2m)^2}$, and its odd terms form $-\sum \frac{1}{2m-1}$, which diverges. A convergent series plus a divergent one diverges.) ■

23. Ratio Test

(b) $\sum_{n=0}^{\infty} \frac{n!}{n^n}$

Solution. (Start at $n = 1$, since 0^0 is undefined.) With $a_n = \frac{n!}{n^n}$,

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \left(\frac{n}{n+1} \right)^n = \frac{1}{\left(1 + \frac{1}{n}\right)^n} \rightarrow \frac{1}{e} < 1.$$

By the Ratio Test it converges. ■

(c) $\sum_{n=0}^{\infty} \frac{(2n+1)!}{3^n}$

Solution.

$$\frac{a_{n+1}}{a_n} = \frac{(2n+3)!}{3^{n+1}} \cdot \frac{3^n}{(2n+1)!} = \frac{(2n+3)(2n+2)}{3} \rightarrow \infty.$$

By the Ratio Test it diverges. ■

(d) $\sum_{n=1}^{\infty} \frac{n^2 4^n}{n!}$

Solution.

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^2 4^{n+1}}{(n+1)!} \cdot \frac{n!}{n^2 4^n} = \frac{4(n+1)}{n^2} \rightarrow 0 < 1.$$

By the Ratio Test it converges. ■

24. Root Test

(a) $\sum_{n=1}^{\infty} \left(\frac{2n+1}{3n-1}\right)^n$

Solution.

$$\sqrt[n]{a_n} = \frac{2n+1}{3n-1} \rightarrow \frac{2}{3} < 1.$$

By the Root Test it converges. ■

(b) $\sum_{n=1}^{\infty} \frac{n^n}{2^{n^2}}$

Solution.

$$\sqrt[n]{a_n} = \frac{n}{2^n} \rightarrow 0 < 1$$

(by L'Hopital, $\lim_{x \rightarrow \infty} \frac{x}{2^x} = \lim_{x \rightarrow \infty} \frac{1}{2^x \ln 2} = 0$). By the Root Test it converges. ■

(c) Show the Root Test is inconclusive for $\sum \frac{1}{n}$ and $\sum \frac{1}{n^2}$. Which converges?

Solution. First, $\ln(n^{1/n}) = \frac{\ln n}{n} \rightarrow 0$, so $n^{1/n} \rightarrow 1$. Then

$$\sqrt[n]{\frac{1}{n}} = \frac{1}{n^{1/n}} \rightarrow 1, \quad \sqrt[n]{\frac{1}{n^2}} = \frac{1}{(n^{1/n})^2} \rightarrow 1.$$

Both limits equal 1, where the Root Test says nothing. Yet $\sum \frac{1}{n^2}$ converges ($p = 2$) and $\sum \frac{1}{n}$ diverges ($p = 1$). ■

(d) $\sum_{n=1}^{\infty} \frac{2^n}{n^5}$: Root Test? Ratio Test?

Solution. Both work. Root: $\sqrt[n]{a_n} = \frac{2}{(n^{1/n})^5} \rightarrow 2 > 1$. Ratio: $\frac{a_{n+1}}{a_n} = 2 \left(\frac{n}{n+1}\right)^5 \rightarrow 2 > 1$.

Either way the series diverges. ■

25. Absolute/Conditional

(a) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

Solution. $\sum \left| \frac{(-1)^n}{n} \right| = \sum \frac{1}{n}$ diverges, so it is not absolutely convergent. With $b_n = \frac{1}{n}$ positive, decreasing, and $b_n \rightarrow 0$, the AST shows it converges. So it converges conditionally. ■

(c) $\sum_{k=2}^{\infty} \frac{\cos(\pi k)}{\ln k}$

Solution. This is $\sum \frac{(-1)^k}{\ln k}$. For $k \geq 2$, $\ln k < k$, so $\frac{1}{\ln k} > \frac{1}{k}$, and $\sum \frac{1}{\ln k}$ diverges by the DCT. The AST applies as in 22(c), so it converges conditionally. ■

(d) $\sum_{n=1}^{\infty} \frac{(-1)^n n^3}{2^n}$

Solution. Apply the Ratio Test to $|a_n| = \frac{n^3}{2^n}$:

$$\frac{|a_{n+1}|}{|a_n|} = \frac{1}{2} \left(\frac{n+1}{n} \right)^3 \rightarrow \frac{1}{2} < 1.$$

So $\sum |a_n|$ converges, and the series converges absolutely. ■

(e) $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n^2+1}$

Solution. Absolute values: $\frac{n}{n^2+1}$ compared with $\frac{1}{n}$ gives $\lim_{n \rightarrow \infty} \frac{\frac{n}{n^2+1}}{\frac{1}{n}} = 1$, so $\sum \frac{n}{n^2+1}$ diverges by the LCT. AST: $b_n = \frac{n}{n^2+1} \rightarrow 0$, and $f(x) = \frac{x}{x^2+1}$ has $f'(x) = \frac{1-x^2}{(x^2+1)^2} \leq 0$ for $x \geq 1$, so b_n is decreasing. The series converges conditionally. ■

(f) Suppose $\sum a_n$ converges to 2. What is $\lim a_n$? Must $\sum |a_n|$ converge?

Solution. Since the series converges, $\lim_{n \rightarrow \infty} a_n = 0$. No, $\sum |a_n|$ need not converge. The series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges to $\ln 2$ but not absolutely. Multiplying every term by $\frac{2}{\ln 2}$ gives a series that converges to 2 whose absolute values still form a multiple of the harmonic series. ■

26. Radius of Convergence (ROC) and Interval of Convergence (IOC)

(a) $\sum_{n=0}^{\infty} n^n x^n$

Solution. For $x \neq 0$, the Ratio Test gives

$$\left| \frac{(n+1)^{n+1} x^{n+1}}{n^n x^n} \right| = (n+1) \left(1 + \frac{1}{n} \right)^n |x| \rightarrow \infty.$$

So the series diverges for every $x \neq 0$. ROC $R = 0$, IOC $\{0\}$. ■

(e) $\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{3^n (2n+1)}$

Solution. Ratio Test:

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{x^2}{3} \cdot \frac{2n+1}{2n+3} \rightarrow \frac{x^2}{3}.$$

It converges when $x^2 < 3$, so $R = \sqrt{3}$. At $x = \pm\sqrt{3}$, $x^{2n} = 3^n$ and the series is $\sum \frac{(-1)^n}{2n+1}$, which converges by the AST. IOC $[-\sqrt{3}, \sqrt{3}]$. ■

(f) $\sum_{n=1}^{\infty} \frac{(x+2)^n}{n 4^n}$

Solution. Ratio Test: $\left| \frac{a_{n+1}}{a_n} \right| = \frac{|x+2|}{4} \cdot \frac{n}{n+1} \rightarrow \frac{|x+2|}{4}$. So it converges for $|x+2| < 4$: center -2 , $R = 4$. At $x = 2$: $\sum \frac{1}{n}$ diverges. At $x = -6$: $\sum \frac{(-1)^n}{n}$ converges by the AST. IOC $[-6, 2)$. ■

(g) Center, ROC, IOC: $\sum_{n=1}^{\infty} \frac{2^n (x-3)^n}{n^2}$

Solution. The center is 3. Ratio Test: $\left| \frac{a_{n+1}}{a_n} \right| = 2|x-3| \left(\frac{n}{n+1} \right)^2 \rightarrow 2|x-3|$, so $R = \frac{1}{2}$. At $x = \frac{7}{2}$: $\sum \frac{1}{n^2}$ converges. At $x = \frac{5}{2}$: $\sum \frac{(-1)^n}{n^2}$ converges absolutely. IOC $[\frac{5}{2}, \frac{7}{2}]$. ■

27. Power series from Geometric Series

- (b) $g(x) = \frac{1}{1+x}$, $a = 1$. State the ROC and IOC.

Solution. Write it in terms of $x - 1$: $1 + x = 2 + (x - 1)$, so

$$\frac{1}{1+x} = \frac{1}{2} \cdot \frac{1}{1 - (-\frac{x-1}{2})} = \sum_{n=0}^{\infty} \frac{(-1)^n (x-1)^n}{2^{n+1}}.$$

It converges when $|\frac{x-1}{2}| < 1$, so $R = 2$ and IOC $(-1, 3)$ (at both endpoints the terms do not go to 0). ■

- (f) $g(x) = \ln(x)$, $a = 1$. State the ROC.

Solution. $\ln x = \int_1^x \frac{dt}{t}$, and $\frac{1}{t} = \frac{1}{1+(t-1)} = \sum_{n=0}^{\infty} (-1)^n (t-1)^n$ for $|t-1| < 1$. Integrate term by term from 1 to x :

$$\ln x = \sum_{n=0}^{\infty} \frac{(-1)^n (x-1)^{n+1}}{n+1} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} (x-1)^n}{n}, \quad R = 1.$$

(Centered at 0 this is impossible, since $\ln x$ is undefined at 0.) ■

- (i) Find the function that $\sum_{n=2}^{\infty} 2n(n-1)x^n$ represents.

Solution. Differentiate $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ twice: $\frac{2}{(1-x)^3} = \sum_{n=2}^{\infty} n(n-1)x^{n-2}$ for $|x| < 1$. Multiply by $2x^2$:

$$\sum_{n=2}^{\infty} 2n(n-1)x^n = \frac{4x^2}{(1-x)^3}, \quad |x| < 1. \quad \blacksquare$$

- (j) $f(x) = \frac{x}{(1-x^2)^2}$, $a = 0$.

Solution. $\frac{1}{1-x^2} = \sum_{n=0}^{\infty} x^{2n}$ for $|x| < 1$. Differentiate: $\frac{2x}{(1-x^2)^2} = \sum_{n=1}^{\infty} 2nx^{2n-1}$. Divide by 2:

$$\frac{x}{(1-x^2)^2} = \sum_{n=1}^{\infty} nx^{2n-1}, \quad R = 1. \quad \blacksquare$$

- (k) Find the exact value of $\sum_{n=1}^{\infty} \frac{n}{3^n}$.

Solution. Differentiate the geometric series and multiply by x : $\sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2}$ for $|x| < 1$. At $x = \frac{1}{3}$:

$$\sum_{n=1}^{\infty} \frac{n}{3^n} = \frac{1/3}{(2/3)^2} = \frac{3}{4}. \quad \blacksquare$$

28. Taylor's Theorem

- (b) Find the Maclaurin series for $\cosh(x)$ and $\sinh(x)$.

Solution. Add and subtract $e^x = \sum \frac{x^n}{n!}$ and $e^{-x} = \sum \frac{(-1)^n x^n}{n!}$. The odd terms cancel in the sum and the even terms cancel in the difference.

$$\cosh x = \frac{e^x + e^{-x}}{2} = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}, \quad \sinh x = \frac{e^x - e^{-x}}{2} = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}.$$

■

- (c) Degree 2 Taylor polynomial of $\sqrt{x}$ at $a = 9$.

Solution. $f(9) = 3$, $f'(x) = \frac{1}{2\sqrt{x}}$ so $f'(9) = \frac{1}{6}$, and $f''(x) = -\frac{1}{4x^{3/2}}$ so $f''(9) = -\frac{1}{108}$.

$$T_2(x) = 3 + \frac{x-9}{6} - \frac{(x-9)^2}{216}.$$

■

- (d) Degree 3 Taylor polynomial of $\cos x$ at $a = \frac{\pi}{3}$.

Solution. $f(\frac{\pi}{3}) = \frac{1}{2}$, $f'(\frac{\pi}{3}) = -\sin \frac{\pi}{3} = -\frac{\sqrt{3}}{2}$, $f''(\frac{\pi}{3}) = -\cos \frac{\pi}{3} = -\frac{1}{2}$,
 $f'''(\frac{\pi}{3}) = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$.

$$T_3(x) = \frac{1}{2} - \frac{\sqrt{3}}{2} \left(x - \frac{\pi}{3}\right) - \frac{1}{4} \left(x - \frac{\pi}{3}\right)^2 + \frac{\sqrt{3}}{12} \left(x - \frac{\pi}{3}\right)^3.$$

■

- (e) Maclaurin series for $F(x) = \int_0^x e^{-t^2} dt$ and its ROC.

Solution. Substitute $-t^2$ into $e^u = \sum \frac{u^n}{n!}$, then integrate term by term.

$$e^{-t^2} = \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n}}{n!}, \quad F(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{n! (2n+1)} = x - \frac{x^3}{3} + \frac{x^5}{10} - \cdots.$$

The series for e^u has $R = \infty$, and integrating does not change the radius, so $R = \infty$. ■

- (f) Find the exact value of $1 - \frac{\pi^2}{3^2 \cdot 2!} + \frac{\pi^4}{3^4 \cdot 4!} - \cdots$

Solution. The n th term is $\frac{(-1)^n (\pi/3)^{2n}}{(2n)!}$, which is the cosine series $\cos x = \sum \frac{(-1)^n x^{2n}}{(2n)!}$ at $x = \frac{\pi}{3}$. The sum is $\cos \frac{\pi}{3} = \frac{1}{2}$. ■

- (g) Find the exact value of $\sum_{n=0}^{\infty} \frac{(\ln 3)^n}{n!}$.

Solution. This is $e^x = \sum \frac{x^n}{n!}$ at $x = \ln 3$, so the sum is $e^{\ln 3} = 3$. ■

- (h) Find the exact value of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n 2^n}$.

Solution. $\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$ for $-1 < x \leq 1$. At $x = \frac{1}{2}$ the sum is $\ln \frac{3}{2}$. ■

- (i) Use a Maclaurin series to evaluate $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$.

Solution. $\sin x - x = -\frac{x^3}{3!} + \frac{x^5}{5!} - \dots$, so

$$\frac{\sin x - x}{x^3} = -\frac{1}{6} + \frac{x^2}{120} - \dots \rightarrow -\frac{1}{6}.$$

■

29. Approximations

- (a) Approximate $\int_0^{0.1} \frac{dx}{1+x^4}$ to within 10^{-6} .

Solution. $\frac{1}{1+x^4} = \sum_{n=0}^{\infty} (-1)^n x^{4n}$ for $|x| < 1$, so

$$\int_0^{0.1} \frac{dx}{1+x^4} = \sum_{n=0}^{\infty} \frac{(-1)^n (0.1)^{4n+1}}{4n+1} = 0.1 - \frac{(0.1)^5}{5} + \frac{(0.1)^9}{9} - \dots$$

This is alternating with decreasing terms. By ASE, stopping after two terms has error at most $\frac{(0.1)^9}{9} < 10^{-6}$ (one term is not enough, since $\frac{(0.1)^5}{5} = 2 \times 10^{-6}$). So the integral is about $0.1 - 0.000002 = 0.099998$. ■

- (b) Use $p_3(x)$ to approximate $\tan^{-1}(x)$ near 0, and bound the error at $x = \frac{1}{2}$ with ASE.

Solution. $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$, so $p_3(x) = x - \frac{x^3}{3}$ and $p_3(\frac{1}{2}) = \frac{1}{2} - \frac{1}{24} = \frac{11}{24}$. At $x = \frac{1}{2}$ the series alternates with decreasing terms, so the error is at most the next term:

$$|\tan^{-1}(\frac{1}{2}) - p_3(\frac{1}{2})| \leq \frac{(1/2)^5}{5} = \frac{1}{160}.$$

■

- (c) Bound the same error with Taylor's Inequality, bounding $|f^{(4)}(x)|$ on $[0, \frac{1}{2}]$, where $f(x) = \tan^{-1}(x)$.

Solution. Differentiating four times gives $f^{(4)}(x) = \frac{24x(1-x^2)}{(1+x^2)^4}$. On $[0, \frac{1}{2}]$, $24x \leq 12$, $0 \leq 1 - x^2 \leq 1$, and $(1+x^2)^4 \geq 1$, so $|f^{(4)}(x)| \leq M = 12$. Taylor's Inequality with $n = 3$:

$$|R_3(\frac{1}{2})| \leq \frac{M}{4!} \left(\frac{1}{2}\right)^4 = \frac{12}{24 \cdot 16} = \frac{1}{32}.$$

This bound is valid but weaker than ASE's $\frac{1}{160}$. ■

- (d) $f(x) = \sum_{n=2}^{\infty} 5n(n-1)x^n$: parts (i) to (vi).

Solution. (i) By 27(i), $f(x) = \frac{5}{2} \cdot \frac{4x^2}{(1-x)^3} = \frac{10x^2}{(1-x)^3}$ for $|x| < 1$.

(ii) $f(\frac{1}{2}) = \frac{10/4}{1/8} = 20$.

(iii) The lowest term of the series is $5 \cdot 2 \cdot 1 x^2$, so $p_2(x) = 10x^2$.

(iv) $p_2(\frac{1}{2}) = \frac{10}{4} = \frac{5}{2}$.

(v) $|f(\frac{1}{2}) - p_2(\frac{1}{2})| = 20 - \frac{5}{2} = \frac{35}{2}$.

(vi) The given $f'''(x)$ is a sum of terms increasing on $[0, \frac{1}{2}]$, so its maximum is at $x = \frac{1}{2}$:
 $M = 600 \cdot \frac{1}{4} \cdot 64 + 720 \cdot \frac{1}{2} \cdot 32 + 180 \cdot 16 = 24000$. Then

$$|R_2(\frac{1}{2})| \leq \frac{M}{3!} \left(\frac{1}{2}\right)^3 = \frac{24000}{48} = 500.$$

The bound is true but far from the actual error $\frac{35}{2}$. ■

- (h) How many terms of $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$ are needed to approximate its sum to within 0.001?

Solution. The series alternates and $b_n = \frac{1}{n!}$ decreases to 0. Using the terms $n = 0, \dots, N$, ASE bounds the error by $b_{N+1} = \frac{1}{(N+1)!}$. Since $\frac{1}{6!} = \frac{1}{720} > 0.001$ and $\frac{1}{7!} = \frac{1}{5040} < 0.001$, we need $N + 1 = 7$, that is the 7 terms $n = 0, \dots, 6$. ■

- (i) $f(x) = \ln(1+x)$: find $T_2(x)$ and bound $|f(x) - T_2(x)|$ for $0 \leq x \leq 0.2$.

Solution. $f(0) = 0$, $f'(x) = \frac{1}{1+x}$, $f''(x) = -\frac{1}{(1+x)^2}$, so $T_2(x) = x - \frac{x^2}{2}$. Next, $f'''(x) = \frac{2}{(1+x)^3} \leq 2$ on $[0, 0.2]$, so take $M = 2$:

$$|f(x) - T_2(x)| \leq \frac{M}{3!} x^3 \leq \frac{2}{6} (0.2)^3 = \frac{1}{375} \approx 0.0027.$$

- (j) Approximate $\int_0^1 e^{-x^2} dx$ to within 0.01 (use ASE). ■

Solution. By 28(e) at $x = 1$,

$$\int_0^1 e^{-x^2} dx = 1 - \frac{1}{3} + \frac{1}{10} - \frac{1}{42} + \frac{1}{216} - \dots$$

The terms alternate and decrease. Since $\frac{1}{216} < 0.01$, the first four terms suffice:
 $1 - \frac{1}{3} + \frac{1}{10} - \frac{1}{42} = \frac{26}{35} \approx 0.743$. ■

30. Complex numbers

- (a) Compute $(1+i)^{10}$ in $a+bi$ form.

Solution. $1+i = \sqrt{2} e^{i\pi/4}$. So $(1+i)^{10} = (\sqrt{2})^{10} e^{i10\pi/4} = 32 e^{i5\pi/2} = 32 e^{i\pi/2} = 32i$. ■

- (c) Compute $(i-1)^{12}$ in $a+bi$ form.

Solution. $i-1 = \sqrt{2} e^{i3\pi/4}$. So $(i-1)^{12} = (\sqrt{2})^{12} e^{i9\pi} = 64 e^{i\pi} = -64$. ■

31. Differential equations

- (a) Show that $y = Ce^{x^2}$ solves $y' = 2xy$ for every constant C . Find the solution with $y(0) = 3$.

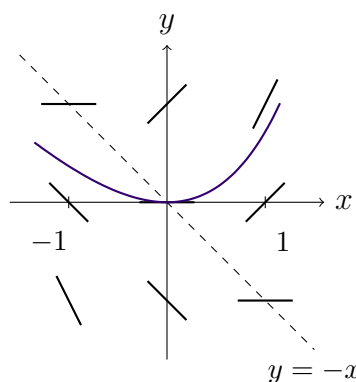
Solution. $y' = C \cdot 2xe^{x^2} = 2x(Ce^{x^2}) = 2xy$. For the initial condition, $y(0) = Ce^0 = C = 3$, so $y = 3e^{x^2}$. ■

- (b) Sketch the direction field of $y' = x + y$ at the nine points with $x, y \in \{-1, 0, 1\}$. Along which line are the slopes 0? Sketch the solution through $(0, 0)$.

Solution. The slope at (x, y) is $x + y$.

	$x = -1$	$x = 0$	$x = 1$
$y = 1$	0	1	2
$y = 0$	-1	0	1
$y = -1$	-2	-1	0

The slopes are 0 exactly on the line $y = -x$. The solution through $(0, 0)$ has slope 0 there. Just to the right the slopes are positive and growing, and just to the left they are negative, so the curve has a minimum at the origin and rises on both sides. (It is $y = e^x - x - 1$.)



■

- (c) Find the equilibrium solutions of $y' = y^2 - 4$ and decide whether nearby solutions move toward or away from each.

Solution. Equilibria are constant solutions, where $y' = 0$: $y = 2$ and $y = -2$. The sign of $y' = (y - 2)(y + 2)$ is positive for $y > 2$, negative for $-2 < y < 2$, and positive for $y < -2$. Near $y = -2$, solutions above it decrease and solutions below it increase, so they move toward $y = -2$. Near $y = 2$, solutions above increase and solutions below decrease, so they move away from $y = 2$.

■

- (d) Euler's method with $h = 0.5$ for $y' = x + y$, $y(0) = 1$, to approximate $y(1)$. Over or under estimate?

Solution. Euler's method steps along the tangent line: $y_{k+1} = y_k + h F(x_k, y_k)$ with $F(x, y) = x + y$.

$$y_1 = 1 + 0.5(0 + 1) = 1.5, \quad y_2 = 1.5 + 0.5(0.5 + 1.5) = 2.5.$$

So $y(1) \approx 2.5$. Differentiating the equation, $y'' = 1 + y' = 1 + x + y > 0$ along the solution here, so the solution is concave up and lies above its tangent lines. Euler's method underestimates. (The exact value is $2e - 2 \approx 3.44$.)

■

- (e) Euler's method with $h = -1$ for $y' = y - x$, $y(1) = 3$, to approximate $y(-1)$.

Solution. $y_{k+1} = y_k + h(y_k - x_k)$ with $h = -1$.

$$y(0) \approx 3 + (-1)(3 - 1) = 1, \quad y(-1) \approx 1 + (-1)(1 - 0) = 0.$$

■

- (f) Solve $\frac{dy}{dx} = xy^2$, $y(0) = 1$. On what interval is the solution defined?

Solution. Separate: $y^{-2} dy = x dx$, so $-\frac{1}{y} = \frac{x^2}{2} + C$. At $x = 0$, $-1 = C$. Then $\frac{1}{y} = 1 - \frac{x^2}{2}$, so

$$y = \frac{2}{2 - x^2}.$$

This blows up where $x^2 = 2$. The interval containing $x = 0$ on which it is defined is $(-\sqrt{2}, \sqrt{2})$.

■

- (g) Solve $\frac{dy}{dx} = e^{x-y}$, $y(0) = \ln 2$.

Solution. $e^{x-y} = e^x e^{-y}$, so separate: $e^y dy = e^x dx$ and $e^y = e^x + C$. At $x = 0$, $2 = 1 + C$, so $C = 1$ and

$$y = \ln(e^x + 1).$$

■

- (h) Solve $\frac{dy}{dt} = 3y - 6$, $y(0) = 5$.

Solution. $\frac{dy}{dt} = 3(y - 2)$. Separate: $\frac{dy}{y-2} = 3 dt$, so $\ln |y - 2| = 3t + C$ and $y - 2 = Ae^{3t}$ for a constant A . At $t = 0$, $A = 3$.

$$y = 2 + 3e^{3t}.$$

■

- (i) Bacteria grow at a rate proportional to size, from 200 cells to 600 in 2 hours. Find $P(t)$ and when $P = 5400$.

Solution. $\frac{dP}{dt} = kP$ gives $P(t) = 200e^{kt}$. From $600 = 200e^{2k}$, $e^{2k} = 3$, so $k = \frac{\ln 3}{2}$ and

$$P(t) = 200 \cdot 3^{t/2}.$$

$5400 = 200 \cdot 3^{t/2}$ means $3^{t/2} = 27 = 3^3$, so $t = 6$ hours.

■

- (j) Logistic: $\frac{dP}{dt} = 0.5P \left(1 - \frac{P}{1000}\right)$, $P(0) = 200$. Carrying capacity, equilibria, growth, and $P(t)$.

Solution. (i) The carrying capacity is $M = 1000$. The equilibria are $P = 0$ and $P = 1000$.

(ii) For $0 < P < 1000$ both factors are positive, so P increases. The rate $0.5P - \frac{0.5P^2}{1000}$ is a downward parabola in P with roots 0 and 1000, so it is largest at the vertex $P = 500$.

(iii) Separate with partial fractions: $\frac{1000}{P(1000-P)} = \frac{1}{P} + \frac{1}{1000-P}$, so $\ln \left| \frac{P}{1000-P} \right| = 0.5t + C$ and $\frac{P}{1000-P} = Be^{0.5t}$ for a constant B . At $t = 0$, $B = \frac{200}{800} = \frac{1}{4}$. Solving for P ,

$$P(t) = \frac{1000}{1 + 4e^{-0.5t}}.$$

$P = 500$ when $4e^{-0.5t} = 1$, that is $t = 2 \ln 4 = 4 \ln 2$.

■